

Amrita School of Engineering, Bengaluru
Department of ECE
B. Tech. AY 2025-2026 (VII Semester)
Synopsis – 23ECE498 & 23EAC498/Project Phase-1

Batch Number/Batch ID:

B001

Name of the Students and Signature (One below the other):

1.Nandakrishna J

2.A Somesh Ayyappan

3.Hari Shankar N

Register number of the students (One below the other):

1. BL.EN.U4ECE23029

2. BL.EN.U4ECE23102

3. BL.EN.U4ECE23118

Name of Guide and Signature:

Dr.Sreeja Kochuvilla

Name of the Domain or TAG:

Robotics & Autonomous Systems

Title of the Project Phase-1:

Design and Development of a Beacon-Based Autonomous Precision Landing System for Unmanned Aerial Vehicles on Static and Moving Platforms.

Project Phase-1 Objective (s) along with the sub-objectives:

To design, evaluate, and develop a reliable UAV platform for autonomous precision landing by selecting appropriate hardware, integrating the flight control system, establishing stable autonomous flight, and preparing the platform for beacon-based landing in subsequent phases.

Sub-objectives

1. Study and analyze the existing UAV platform and identify hardware limitations.
2. Finalize the UAV architecture (frame, motors, ESCs, flight controller, telemetry, power system).
3. Select and procure upgraded components where required.
4. Assemble and integrate the UAV hardware.
5. Configure and calibrate the flight controller using ArduPilot.
6. Perform sensor calibration and PID tuning for stable flight.
7. Validate manual flight and autonomous waypoint navigation.
8. Prepare interfaces for beacon sensors, camera, and companion computer for Phase 2.

References:

1. [Simulation and real-life implementation of UAV autonomous landing system based on object recognition and tracking for safe landing in uncertain environments.pdf](#)
2. [Precision Landing for Low-Maintenance Remote Operations with UAVs.pdf](#)
3. [UWB localization.pdf](#)
4. [Precison landing of drones iisc.pdf](#)
5. [High-Precision Landing on a Moving Platform Based on Drone Vision Using YOLO Algorithm.pdf](#)
6. [A Precision Drone Landing System using Visual and IR.pdf](#)
7. [An_active_beaconbased_target_localization_method_for_Unmanned_Aerial_Vehicles_with_particle_filter.pdf](#)
8. [Robust_Precision_Landing_for_Autonomous_Drones_Combining_Visionbased_and_Infrared_Sensors.pdf](#)

Time Plan of Project Phase-1:

Month	Work to be implemented
July 2026	Literature survey on autonomous precision landing systems. Study the existing drone platform and identify limitations. Finalize project requirements and overall system architecture. Prepare component comparison and shortlist hardware upgrades.
August 2026	Finalize the hardware platform. Procure required components (frame, motors, ESC, flight controller, telemetry, battery, GPS, camera, etc.). Design power distribution and communication architecture.
September 2026	Assemble the UAV. Integrate Pixhawk/ArduPilot, power module, GPS, telemetry, receiver, and propulsion system. Configure firmware, perform sensor calibration, ESC calibration, and communication testing.
October 2026	Conduct ground testing and flight testing. Perform PID tuning, vibration analysis, EKF verification, and autonomous waypoint flight validation. Resolve stability and latency issues.
November 2026	Validate the stable UAV platform. Document hardware specifications and flight performance. Integrate preliminary interfaces for beacon receiver, global shutter camera, and companion computer to prepare for autonomous precision landing in Phase 2.